RAG PDF Chatbot

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Ask a Question

What is BERT?

Answer

Based on the provided context, BERT is a transformer-based language model developed by Google that is designed for deep understanding of text. Unlike traditional models that read text in one direction, BERT reads sentences bidirectionally (left and right context together), allowing it to better understand the meaning of words based on surrounding context. It is mainly used for Natural Language Understanding (NLU) tasks such as classification, question answering, and sentence similarity.

Retrieved Context

1. Definition
   BERT is a transformer-based language model developed by Google that is designed for deep understanding of text. Unlike traditional models that read text in one direction, BERT reads sentences bidirectionally (left and right context together), allowing it to better understand the meaning of words based on surrounding context. It is mainly used for Natural Language Understanding (NLU) tasks such as classification, question answering, and sentence similarity.

2. Architecture / Diagram
   Encoder only from transformer

3. Working (Step-by-Step)

4. Input text is converted into tokens and special tokens like [CLS] (classification) and [SEP] (separator) are added.

5. Each token is converted into embeddings (word + position + segment embeddings). 3. The input passes through multiple transformer encoder layers.

6. Each layer uses self-attention to understand relationships between words.

7. The model learns context by predicting masked words (MLM).

Decoder: ⊠ Takes encoder output + previous outputs
⊠ Generates final output (word-by-word)
Working Flow: Input → Encoder → Encoded Representation → Decoder → Output

4. Advantages ⊠ Handles long-range dependencies effectively
   ⊠ Highly parallel → faster training
   ⊠ Better performance than RNN/LSTM ⊠ Forms base of modern models (GPT, BERT)

5. Disadvantages ⊠ Requires large computational resources
   ⊠ Needs large datasets for training
   ⊠ Complex architecture
   ⊠ Memory intensive

6. Applications ⊠ Machine translation (Google Translate)
   ⊠ Chatbots and AI assistants
   ⊠ Text summarization
   ⊠ Question answering systems
   ⊠ Code generation

2.2 Evolution of Transformer Models 2.2.1 BERT (Bidirectional Encoder Representations from Transformers)

1. Definition
   BERT is a transformer-based language model developed by Google that is designed for deep understanding of text. Unlike traditional models that read text in one direction, BERT reads

2. Generates output step-by-step (autoregressive generation).

3. Optimizations improve speed and performance compared to traditional models.

4. Advantages ⊠ High performance with lower computational cost
   ⊠ Faster and more efficient than many large models

 Suitable for real-time applications
 Good balance between size and accuracy

5. Disadvantages Newer model → less mature ecosystem
 Limited documentation compared to older models May require tuning for specific tasks
 Not as widely adopted as GPT/BERT

6. Applications Chatbots and AI assistants
 Text generation
 Real-time AI systems
 Research and development
 Content creation
Feature BERT GPT (2/3/4) T5 LLaMA Claude Mistral Developed By Google OpenAI Google Meta Anthropic Mistral AI Architecture Encoder-only Decoder- only Encoder– Decoder Decoder- only Decoder- based Optimized Transformer Main Purpose Understanding text Generating text All tasks as text Efficient

7. Applications Chatbots and conversational AI
 Text generation
 Research in NLP
 Educational tools
 Content creation

2.2.5 Claude

1. Definition Claude is a transformer-based conversational AI model developed by Anthropic. It is designed to generate helpful, safe, and reliable responses in natural language. Unlike many earlier models, Claude focuses strongly on AI safety and alignment, meaning it aims to follow human values and avoid harmful or misleading outputs. It is widely used in chat-based applications and AI assistants.

2. Architecture (Not Required) (You can skip diagram — similar to GPT-style transformer models)

3. Working (Step-by-Step)

4. User provides an input prompt.

5. Input is converted into tokens.

6. The model processes tokens using a transformer architecture.

7. It generates responses word-by-word (autoregressive).

8. Uses human feedback (RLHF-like methods) to improve safety and quality.

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